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The $D\,^1\Sigma^+$ state of ^{7}LiH

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The ^{7}LiH $D\,^1\Sigma^+$ excited electronic state has been observed for the first time by a pulsed optical–optical double resonance fluorescence depletion spectroscopic technique. Several rovibrational levels of the $A\,^1\Sigma^+$ and $B\,^1\Pi$ electronic states have been used as the intermediate states. Among 22 vibrational levels, 128 rovibrational levels have been observed. The spectral assignment is identified by the observed rotational structures and term values and by a comparison between the derived vibrational and rotational constants with the theoretical values. An *ab initio* adiabatic potential energy curve for the $D\,^1\Sigma^+$ state and relevant $D\,^1\Sigma^+-A\,^1\Sigma^+$ and $D\,^1\Sigma^+-B\,^1\Pi$ transition moment functions are presented. An overall measure of three different aspects, the energetics, the spectral line shape, and the transition probability, strongly favors the argument that the vibronic approach is a better approach for characterizing the excited $D\,^1\Sigma^+$ electronic state. © *2000 American Institute of Physics.* [S0021-9606(00)01226-5]

I. INTRODUCTION

Lithium hydride, the simplest neutral heteronuclear molecule, has been the object of intensive theoretical and spectroscopic interest since the 1930s. A comprehensive and critical review for this molecule has been compiled by Stwalley and Zemke.[1] Up to now only the lowest four singlet electronic states, $X\,^1\Sigma^+$, $A\,^1\Sigma^+$, $B\,^1\Pi$, and $C\,^1\Sigma^+$, have been identified and well characterized by conventional spectroscopy[2–7] and by laser spectroscopy,[8–18] although some other electronic states have been predicted theoretically.[19–25] The potential energy curves of four observed singlet electronic states are shown in Fig. 1, in which an *ab initio* $D\,^1\Sigma^+$ potential calculated in this work (see Sec. III below) and the Li^+H^- ionic curve (including only the R^{-1} term) are also included.

In an effective core potential calculation, Boutalib and Gadéa[24] reported that the potential energy curves for highly excited electronic states of LiH would become complex due to the avoided crossings between the valence and Rydberg atomic configurations on the one hand, and between the ionic and neutral configurations on the other hand. The first intriguing double-well $C\,^1\Sigma^+$ electronic state has been recently investigated by Luh *et al.*[17,18] using a pulsed optical–optical double resonance (OODR) fluorescence depletion spectroscopic technique. A theoretical spectroscopic study, which assumes complete break down of the Born–Oppenheimer approximation for this state, has been reported by Gemperle and Gadéa.[26] The calculated nonradiative lifetimes for all $C\,^1\Sigma^+$ vibrational levels varied over the range 2 ns–0.7 ms, and they appeared not short enough to be reflected in the OODR spectral linewidth. In a very recent vibronic calculation for the double-well $D\,^1\Sigma^+$ state, Gemperle and Gadéa[27] have argued that: (i) the adiabatic Born–Oppenheimer approximation had completely broken down, (ii) the vibronic shift was quite large, and (iii) the calculated nonradiative lifetimes for the lowest 26 levels varied over the range 0.07 ps–60 ns, corresponding to the resonance widths [half-width at half-maximum (HWHM)] over the range 4×10^{-5}–$38\,\mathrm{cm}^{-1}$. These predictions can be tested through OODR spectral linewidths, which will be broadened for transitions that originate from short-lived molecular states.

In this work we present the first OODR results on the $D\,^1\Sigma^+$ state of ^{7}LiH. Among 22 vibrational levels, 128 rovibrational levels have been observed. The spectral assignment is identified by the observed rotational structure and term values and by a comparison between the derived vibrational and rotational constants and their theoretical values. An *ab initio* adiabatic potential energy curve for the $D\,^1\Sigma^+$ state and relevant $D\,^1\Sigma^+-A\,^1\Sigma^+$ and $D\,^1\Sigma^+-B\,^1\Pi$ transition moment functions are calculated and presented. An overall measure of three different aspects, the energetics, the spectral line shape, and the transition probability, strongly favors the argument that the vibronic approach, which assumes a complete breakdown of the Born–Oppenheimer approximation, is a better approach for characterizing the excited $D\,^1\Sigma^+$ electronic state.

II. EXPERIMENT

The experimental setup is similar to the one reported previously.[18] Below, we describe only the significant fea-

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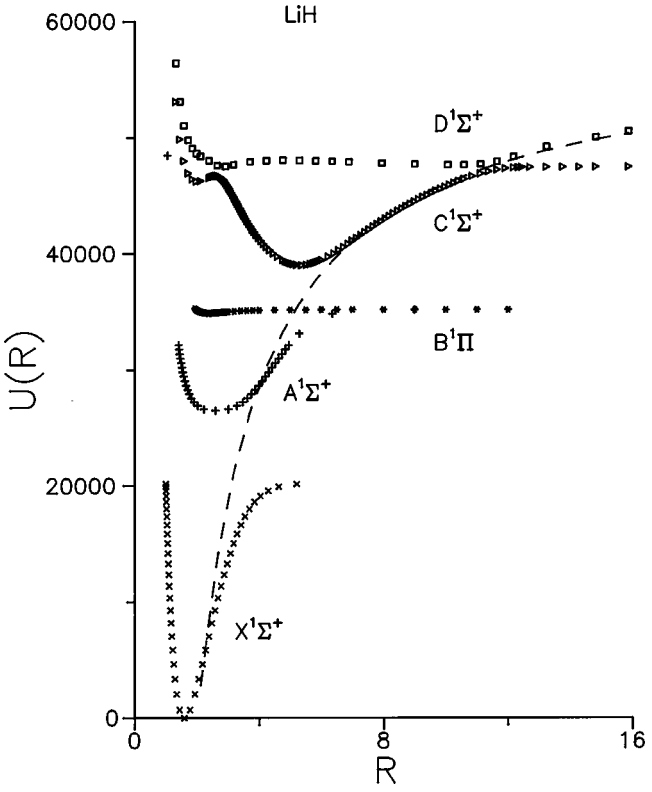


FIG. 1. Potential energy curves of ^{7}LiH: three IPA potentials for the $X\,^1\Sigma^+$, $A\,^1\Sigma^+$, and $B\,^1\Pi$ states (from Ref. 1), an effective hybrid $C\,^1\Sigma^+$ potential (from Ref. 18), and an *ab initio* $D\,^1\Sigma^+$ potential calculated in this work. The dashed line represents the Li^+H^- ionic curve. $U(R)$ in cm^{-1} and R in Å.

tures. The pump laser system was a Nd:YAG laser (Lumonics YM 600) pumped dye laser (Lumonics HD 500). The O-switched Nd:YAG laser operated at 10 Hz repetition rate with a 7 ns pulse length and 25 mJ pulse energy. The dye laser beam was frequency-doubled by a frequency-doubling system (Inrad autotracker II) to generate an ultraviolet PUMP laser beam for exciting the LiH molecule from the $X\,^1\Sigma^+$ state to the $A\,^1\Sigma^+$ or $B\,^1\Pi$ states. A methanol solution of a dye mixture (R590+R610) was used as the active medium for covering the three B(0-2)-X(0) bands. The PROBE laser was a frequency-doubled/tripled Nd:YAG laser (Quantel BW) pumped dye laser (Lumonics HD500). The Nd:YAG laser repetition rate was 10 Hz, the pulse length 4 ns, and pulse energy 20 mJ. Two laser dyes dissolved in methanol were used: C503 and C540A, for the spectral range 17 390–20 400 cm^{-1} in the D-A-X OODR scheme, and the methanol solutions of a dye mixture (LDS759+LDS821) for the spectral range of 12 195–13 790 cm^{-1} in the D-B-X scheme. The uncertainty of the frequency measurements is $\sim$0.1 cm^{-1} for the D-A case calibrated with a dense I_2 reference spectrum, and $\sim$1.0 cm^{-1} for the D-B case calibrated with a sparse Ar optogalvanic spectrum. The PUMP laser beam and the PROBE laser beam traveled collinearly along the axis of the longer arms of a five-arm crossed heat-pipe oven. The pulse delay between the PROBE laser pulse and the PUMP laser pulse was regulated by two digital delay/pulse generators (Stanford DG535). The heat-pipe oven, containing 10 g of naturally abundant lithium metal (STREM, 3N) in its center, was normally operated under 1 Torr of Ar gas and at the temperature 953 K. The laser induced A-X (or B-X) fluorescence was detected by a photomultiplier (EMI 9863QB) detector after a spectral bandpass filter. The detector signal was fed into a boxcar-averager-gated-integrator system, and the averaged output was then recorded by a personal computer. The OODR fluorescence depletion spectrum was obtained by monitoring the variation of the laser induced A-X (or B-X) fluorescence intensity as the PROBE laser wavelength was scanned, while the PUMP laser was fixed at a particular LiH A-X (or B-X) rovibrational transition.

III. RESULTS AND DISCUSSIONS

A. Calculation

For the PUMP transitions among the $A\,^1\Sigma^+-X\,^1\Sigma^+$ and $B\,^1\Pi-X\,^1\Sigma^+$ electronic transitions, we selected those vibrational bands having known favorable radiative transition probabilities.[28] In order to predict in advance the likely spectral ranges and radiative transition probabilities[29] for the PROBE transitions, we calculated the adiabatic potential energy curves for the $A\,^1\Sigma^+$, $B\,^1\Pi$, and $D\,^1\Sigma^+$ states and the transition moment functions for the $D\,^1\Sigma^+-A\,^1\Sigma^+$ and $D\,^1\Sigma^+-B\,^1\Pi$ electronic transitions. The potential energy curve of the $D\,^1\Sigma^+$ state and the two relevant transition moment functions are listed in Table I. Note that the details of the calculation, already mentioned in previous work,[18,25] will not be further described here.

B. OODR spectra

The OODR spectra were taken by monitoring the laser induced ^{7}LiH $A\,^1\Sigma^+-X\,^1\Sigma^+$ (or $B\,^1\Pi-X\,^1\Sigma^+$) fluorescence intensity at 379 nm through a set of optical filters, having a spectral bandpass [full width at half-maximum (FWHM)] of 36 nm. The rovibrational energy levels, among $A\,^1\Sigma^+(v'=6-9,J'=0-10)$ and $B\,^1\Pi(v'=0,J'=1-6)$, were selected as intermediate states. Note that in the D-B-X OODR scheme, most transitions were observed from the *f*-parity rotational states of the $B\,^1\Pi v'=0$ vibrational level, and no transition was detected out of the rotational states among the $B\,^1\Pi v'=1$ and 2 vibrational levels. Figures 2 and 3 separately present the OODR spectra out of the $A\,^1\Sigma^+(v'=9,J'=3,5)$ and $B\,^1\Pi(v'=0;J'=3e,4f,5e)$ intermediate states. Our spectral results show that excitation from vibrational level leads to two different OODR spectral transitions from each intermediate state among all the $A\,^1\Sigma^+$ and $B\,^1\Pi$ *e*-parity rovibrational levels, but only a single transition is excited from the $B\,^1\Pi$ *f*-parity rovibrational levels. This observed rotational structure indicates that the electronic state involved in the probe laser excitation step is a $^1\Sigma^+$ electronic state. Furthermore, it is an upper $^1\Sigma^+$ electronic state resulting from a normal two-step OODR mode since there is no extra $^1\Sigma^+$ electronic state, lying between the excited $A\,^1\Sigma^+$ state and the ground $X\,^1\Sigma^+$ electronic state, to be operative in a folded OODR mode in the spectral range of 12 195–13 790 cm^{-1} from the $B\,^1\Pi$ state. Note that in Fig. 2 the upward spectral line at 18 311.75 cm^{-1}, which was also observed when the LiH A-X transition was not excited, is attributed to the Li $4d$-$2s$ two-photon atomic transition. An-

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TABLE I. *Ab initio* adiabatic $D\,^1\Sigma^+$ potential energy curve $U(R)$ [hartree] and the transition moment functions, $\mu(R)_{\text{D-A}}$ and $\mu(R)_{\text{D-B}}$ [a.u.], for the $D\,^1\Sigma^+ - A\,^1\Sigma^+$ and $D\,^1\Sigma^+ - B\,^1\Pi$ electronic transitions, respectively. R is the internuclear distance (bohr).

R	$U(R)$[a]	$\mu(R)_{\text{D-A}}$	$\mu(R)_{\text{D-B}}$
1.75	0.145 912	1.0876	-
2.0	0.085 819	1.1198	-
2.5	0.023 694	1.1992	0.2338
2.75	0.008 612	1.2662	0.1187
3.0	−0.000 703	1.3368	0.0086
3.25	−0.006 323	1.4419	0.1515
3.5	−0.009 644	1.5518	0.3060
3.75	−0.011 587	1.6721	0.4687
4.0	−0.012 786	1.7752	0.6700
4.5	−0.014 470	2.0372	1.0368
5.0	−0.016 384	1.9468	1.6099
5.5	−0.016 778	0.8517	2.1787
6.0	−0.016 059	0.3815	2.1908
7.0	−0.014 960	0.1471	2.1119
8.0	−0.014 474	0.0784	2.0424
9.0	−0.014 335	0.0041	1.9935
10.0	−0.014 387	0.1539	1.9567
11.0	−0.014 547	0.4143	1.9263
12.0	−0.014 774	0.7048	1.8983
13.0	−0.015 029	0.9386	1.8736
15.0	−0.015 539	1.2416	1.8286
17.0	−0.015 944	1.4064	1.7622
19.0	−0.016 161	1.4591	-
20.0	−0.016 117	1.3733	0.6262
21.0	−0.015 860	1.1713	-
22.0	−0.014 844	0.5704	-
23.0	−0.012 927	0.0344	-
25.0	−0.009 044	0.2366	-
28.0	−0.005 399	0.0825	-
30.0	−0.002 976	0.1040	-
31.0	−0.001 892	0.1201	-
32.0	−0.000 904	0.1444	-
33.0	−0.000 171	0.1104	-
34.0	−0.000 029	0.0266	-
35.0	−0.000 009	0.0094	-
36.0	−0.000 003	0.0041	-

[a]With respect to the dissociation limit of $\text{Li}(3p)+\text{H}(1s)$.

other three Li two-photon atomic transitions observed are $3s$-$2s$, $3d$-$2s$, and $5d$-$2s$. However, we did not attempt to resolve further their spectral origins.

C. Vibrational and rotational constants

The number of observed OODR transitions is 258, covering a range of 128 rovibrational levels and 22 vibrational levels. The term energy for each observed rovibrational level is obtained by adding together the PROBE frequency and the term energy of the intermediate rovibrational level, which was calculated with known Dunham-type coefficients.[30] Some resultant term energies are plotted in Fig. 4. For each vibrational level with more than two rovibrational levels observed, the vibrational and rotational constants, G_v and B_v, are derived and listed in Table II. Note that the vibrational assignment for each observed vibrational level will be identified and discussed in the following section, and the last observed level was temporarily assigned as $v=32$ based on its energy difference from the level $v=25$ and the likely vibrational spacing. From Table II, one may easily tell that

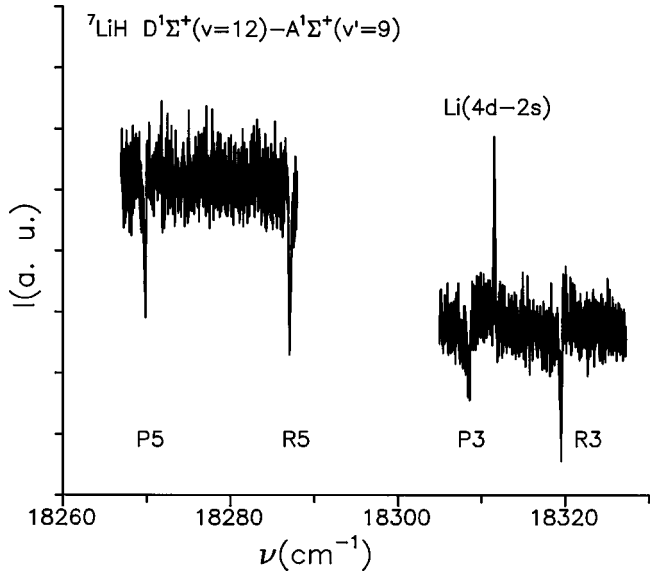


FIG. 2. OODR spectra of ^{7}LiH. The rovibrational levels ($v'=9$, $J'=3,5$) of the $A\,^1\Sigma^+$ state were selected as intermediate levels.

(i) the listed G_v values, covering the energy range 47 667–49 431 cm^{-1} above the potential minimum of the ground $X\,^1\Sigma^+$ state, lie energetically within two *ab initio* ranges, one 47 518–51 213 cm^{-1} (this work) and the other 47 363–51 100 cm^{-1} (see Ref. 27), for the $D\,^1\Sigma^+$ state; and (ii) the derived rotational constants B_v,0.18–0.95 cm^{-1}, are close to the theoretical B_v constants, whose values were calculated to be 2.28 and 1.84 cm^{-1} for the inner well $v=1$ and 6 levels, 0.19–0.38 cm^{-1} for the outer well $v=0$, 2–5, and 7–8 levels, and 0.67–86 cm^{-1} for the $v=9-25$ levels above the intermediate barrier. Based on these two points, the electronic state involved is indeed the $D\,^1\Sigma^+$ state of interest. Note that theoretical molecular constants (B_v and G_v) were calculated by using an eigenvalue program[31] with the *ab initio* $D\,^1\Sigma^+$ potential presented in Table I. This adiabatic po-

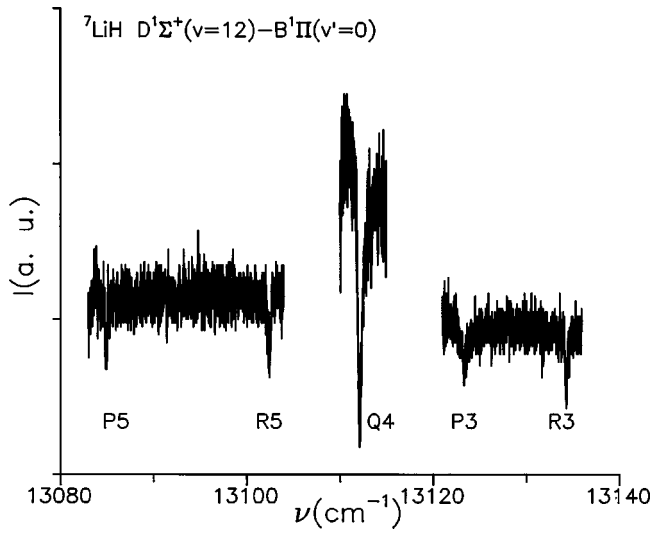


FIG. 3. OODR spectra of ^{7}LiH. The rovibrational levels ($v'=0$, $J'=3e,4f,5e$) of the $B\,^1\Pi$ state were selected as intermediate levels.

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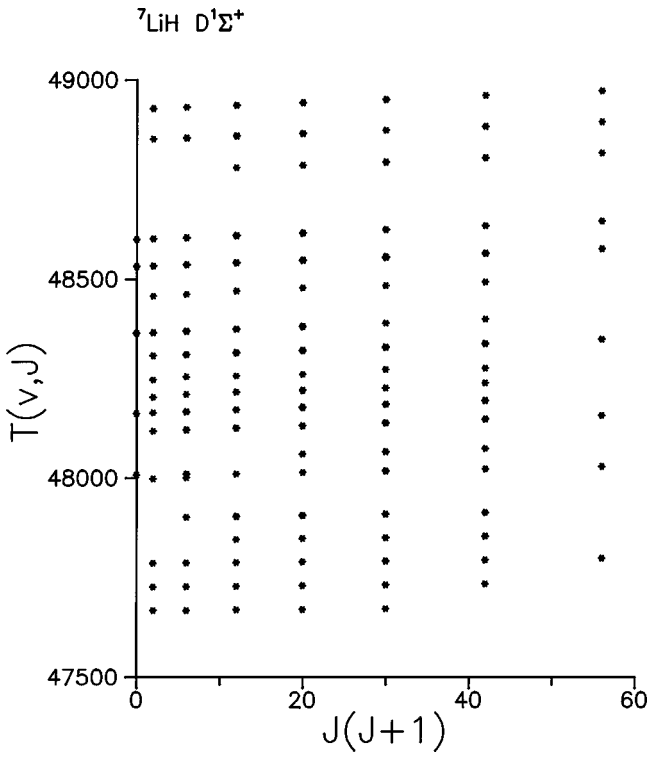


FIG. 4. Plot of observed term energies $T(v,J)$ [in cm^{-1}] of the 125 observed rovibrational levels, among the 22 vibrational levels of the ^{7}LiH $D\,^1\Sigma^+$ state, with respect to the $J(J+1)$ product.

tential and its lowest 16 vibrational ($J=0$) levels are plotted in Fig. 5, which reveals its peculiar double well.

D. The vibronic approach

In a recent theoretical work, Gemperle and Gadéa[27] (GG) reported that: (i) the adiabatic Born–Oppenheimer approach, an approach adequate for the known $X\,^1\Sigma^+$, $A\,^1\Sigma^+$, $B\,^1\Pi$, and $C\,^1\Sigma^+$ states, had completely broken down for the $D\,^1\Sigma^+$ electronic state; (ii) the vibronic shift was fairly large; and (iii) the resonance widths varied over the range 4×10^{-5}–$38\,cm^{-1}$. Their work implied that if one would like to characterize well the LiH $D\,^1\Sigma^+$ state, then the vibronic approach, which assumes the breakdown of the Born–Oppenheimer approximation, would be better. Such an implication may become evident via three different aspects: first the energetics, second the OODR spectral line shape (linewidth), and finally the OODR transition probability. First, one may compare observed vibrational energies (G_v) and vibrational spacings ($\Delta G_{v+1/2}=G_{v+1}-G_v$, here denoted as ΔG_v for brevity) with their corresponding theoretical results. Observed vibrational ($G_{v,O}$) energies and the corresponding relative energy differences ($\delta_{A\text{-}O}, \delta_{v\text{-}O}$) for the likely adiabatic and vibronic levels are listed in Table III, in which every observed vibrational spacing as well as the adiabatic and vibronic ones ($\Delta G_{v,O}$, $\Delta G_{v,A}$, and $\Delta G_{v,V}$) are included for comparison. For an easy comparison, the vibrational energy levels are listed in terms of the inner well levels, the outer well levels, and the levels above the intermediate barrier, located at ~4.83 Å and ~48 060 cm^{-1} (this work), and the likely second inner well levels, $v=6$ (this work) and 7 (GG), are listed twice in Table III. For the 20 observed vibrational levels, the relative energy differences are found to vary over the range -9–$+72$ cm^{-1} for the adiabatic vibrational levels calculated from the potential presented in Table I, and -36–$+9$ cm^{-1} and -7–$+13$ cm^{-1}, respectively, for the GG adiabatic and vibronic ones. The above energy comparison reveals that (i) the adiabatic potential presented in Table I is slightly narrower and at higher

TABLE II. Vibrational and rotational constants, G_v and B_v, (in cm^{-1}) for 21 observed vibrational levels of the ^{7}LiH $D\,^1\Sigma^+$ state. σ represents the uncertainty for overall fitting, and A and B the $A\,^1\Sigma^+$ and $B\,^1\Pi$ electronic states, whose rovibrational levels were used as intermediate states. The parenthesized number represents one standard deviation.

v	G_v	B_v	σ	
0	476 67.34(19)	0.1837(115)	0.26	*B*
2	477 26.61(11)	0.1963(49)	0.17	*B*
3	477 83.42(15)	0.2787(38)	0.08	*A, B*
4	478 43.16(33)	0.2842(119)	0.27	*A, B*
5	479 00.35(11)	0.3133(109)	0.08	*A*
7	479 96.37(191)	0.9488(1578)	2.14	*B*
8	480 07.57(22)	0.3024(187)	0.30	*A*
9	480 47.88(55)	0.6338(174)	0.27	*A*
11	481 16.74(6)	0.7489(27)	0.11	*A, B*
12	481 62.21(3)	0.7805(30)	0.04	*A, B*
13	482 03.32(193)	0.8596(763)	2.37	*A, B*
14	482 48.10(184)	0.7474(790)	2.69	*B*
15	483 06.18(8)	0.7910(52)	0.04	*A, B*
16	483 64.24(3)	0.8274(36)	0.06	*A, B*
18	484 57.87(162)	0.8881(697)	2.37	*B*
19	485 31.48(1)	0.8212(13)	0.03	*A, B*
20	485 98.60(1)	0.8363(14)	0.03	*A, B*
23	487 69.57(43)	0.8409(120)	0.42	*A*
24	488 49.49(7)	0.8342(63)	0.11	*A*
25	489 26.82(2)	0.8252(19)	0.03	*A*
32[a]	494 30.57(7)	0.7838(61)	0.09	*A*

[a]A temporary assignment in term of energy difference and likely vibrational spacing.

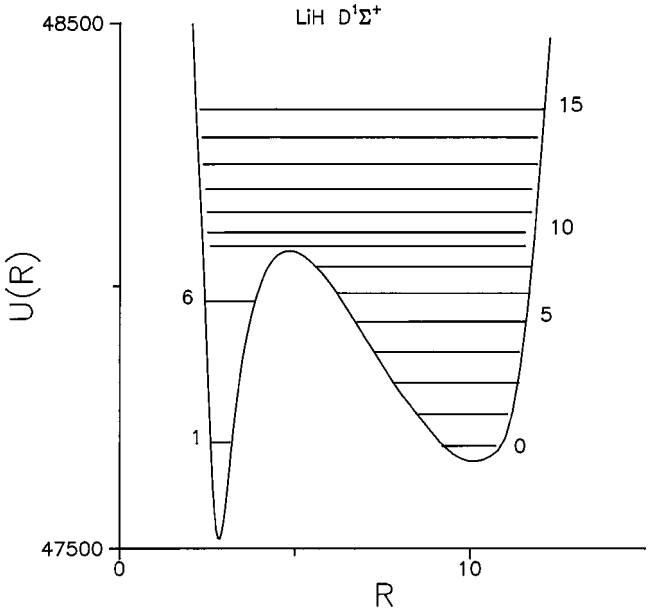


FIG. 5. The adiabatic potential energy curve of the ^{7}LiH $D\,^1\Sigma^+$ state and its lowest 16 vibrational levels. $U(R)$ in cm^{-1} and R in Å.

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TABLE III. Vibrational energies G_v and vibrational energy spacings ΔG_v. The v value represent the vibration quantum number, $G_{v,O}$ the observed vibrational energy, $\delta_{A\text{-}O}$ and $\delta_{V\text{-}O}$ the relative energy differences of the adiabatic and vibronic vibrational energies relative to the observed one, and $\Delta G_{v,O}$, $\Delta G_{v,A}$ and $\Delta G_{v,V}$ the observed, adiabatic, and vibronic vibrational energy spacings, respectively.

	This work		GG		This work		GG[c]	
v	$G_{v,O}$	$\delta_{A\text{-}O}$[a]	$\delta_{A\text{-}O}$[b]	$\delta_{V\text{-}O}$[b]	$\Delta G_{v,O}$	$\Delta G_{v,A}$	$\Delta G_{v,A}$	$\Delta G_{v,V}$
(Primarily inner well)								
1	-	-	-	-	-	267.90	272.44	227.32
6	d	e	-	-	-	15.89	2.45	39.36
7	47 996.37	−9.76	−35.40[e]	6.80[e]	11.20	50.51	48.95	9.81
(Primarily outer-well)								
0	47 667.34	28.13	8.09	9.43	59.27	60.91	57.46	58.72
2	47 726.61	29.77	6.28	8.88	56.81	59.98	57.81	58.12
3	47 783.42	22.94	7.28	10.19	59.74	58.40	57.28	56.95
4	47 843.16	31.59	4.82	9.40	57.19	57.55	56.20	55.21
5	47 900.35	31.95	3.84	7.41	-	38.43	54.34	56.06
6	d	e	-	-	-	15.89	2.45	39.36
7	47 996.37	−9.76	−35.40[e]	6.80[e]	11.20	50.51	48.95	9.81
8	48 007.57	29.56	2.35	5.42	40.31	38.83	45.33	47.25
9	48 047.88	28.08[f]	7.37	12.36	-	26.35	28.77	36.11
(Above the intermediate barrier)								
10	-	-	-	-	-	38.52	30.05	28.63
11	48 116.74	24.09	−2.67	8.23	45.47	44.08	40.70	43.60
12	48 162.21	22.70	−7.44	6.36	41.11	47.44	43.84	47.97
13	48 203.32	29.03	−4.71	13.22	44.78	50.74	46.78	42.83
14	48 248.10	35.00	−2.71	11.27	58.08	53.50	49.43	51.20
15	48 306.18	30.42	−11.36	4.39	58.06	56.10	51.76	55.29
16	48 364.24	28.46	−17.66	1.61	-	58.39	53.85	62.27
17	-	-	-	-	-	60.44	55.73	42.92
18	48 457.87	53.66	−1.70	13.18	73.61	62.39	57.48	60.94
19	48 531.48	42.44	−17.83	0.51	67.12	64.17	59.14	66.09
20	48 598.60	39.49	−25.80	−0.52	-	65.94	60.73	70.57
21	-	-	-	-	-	67.61	62.21	71.04
22	-	-	-	-	-	69.21	63.55	22.88
23	48 769.57	71.28	−10.29	−7.00	79.92	70.65	64.80	82.87
24	48 849.49	62.01	−25.41	−4.05	77.33	71.90	65.89	76.54
25	48 926.82	56.60	−26.85	−4.83				

[a]Calculated with the potential energy curve in Table I by taking into account the calculated Li $3p-2s$ energy difference, 30 924.85 cm^{-1}, and the dissociation energy (Ref. 1), $D_e=20\,287.7$ cm^{-1}, of the ^{7}LiH X $^1\Sigma^+$ state.
[b]Calculated from the tabulated values of Ref. 27 by adding together the above D_e value.
[c]Calculated directly from the tabulated values of Ref. 27.
[d]One term value, 47 971.07 cm^{-1}, observed from the intermediate state $A(8,6)$.
[e]The second inner well level.
[f]Above the intermediate barrier.

energy than the real potential, but the GG adiabatic potential appears broader and at lower energy; and (ii) the vibronic levels are relatively close to the observed vibrational levels indicating that the vibronic representation is a better one.

From Table III one can also easily tell that the four adiabatic and vibronic vibrational spacings, among the lowest five outer well levels, match quite well with the first four observed spacings. Thus combining with the consistency between derived rotational constants and theoretical ones, one can surely identify the lowest observed level as the outer well $v=0$ level and the next four observed vibrational levels as the outer well $v=2-5$ levels. For the 25 calculated vibrational spacings, the two adiabatic sets ($\Delta G_{v,A}$) vary similarly, but the vibronic set ($\Delta G_{v,V}$) varies in a different way. In the vibronic approach, the two inner well vibrational levels are each dramatically shifted ($\sim$87 and $\sim$42 cm^{-1}; see also Table I of Ref. 27), and in turn the vibronic spacings ($\Delta G_{v,V}$) become different from their respective adiabatic ones. Particularly at $v=7$, the adiabatic ($\Delta G_{v,A}$) values are 50.51 and 48.95 cm^{-1}, but the vibronic ($\Delta G_{v,V}$) value is 9.81 cm^{-1}, which is closer to the observed ($\Delta G_{v,O}$) value of 11.20 cm^{-1}. At $v=23$ and 24, the vibronic ($\Delta G_{v,V}$) values also match better with the observed ($\Delta G_{v,O}$) values. In summary, the above comparison of the vibrational spacings also favors the argument that the vibronic approach is better.

Second, the results of the vibronic calculation[27] revealed that the vibronic resonance widths for different vibrational levels varied quite irregularly, and in turn the OODR spectral line shapes were expected to change accordingly. For visualizing such a variation, two $D\,^1\Sigma^+-A\,^1\Sigma^+$ and two $D\,^1\Sigma^+-B\,^1\Pi$ OODR spectra are separately presented in Figs. 6 and 7. Furthermore, 19 observed (not deconvoluted) spectral linewidths (HWHM) and 26 vibronic resonance widths[27] are plotted in Fig. 8. These three figures together reveal that: (i) the observed spectral line shape changes as the vibrational level varies; (ii) those observed vibrational

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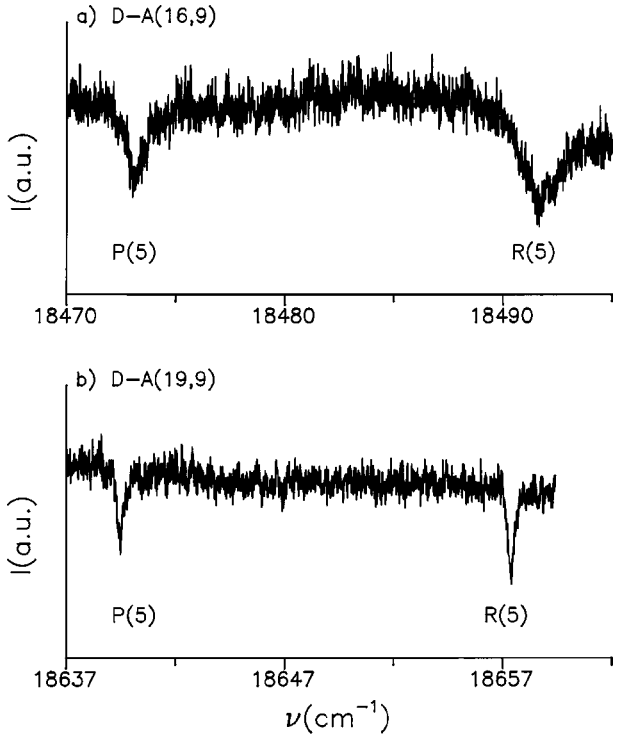


FIG. 6. OODR spectra. (a) Two broad spectral lines represent the D-A(16, 9), P(5), and R(5) transitions, and (b) two narrow ones the D-A(19,9), P(5), and R(5) transitions.

levels having a broad spectral linewidth correspond rather well to those levels having a broad vibronic resonance width; and (iii) in sum, the vibronic approach is again better.

Finally, one may compare the relative intensities of observed OODR spectral transitions with theoretical values. Here the relevant adiabatic potential and transition moment functions for the *D* state were already presented in Table I.

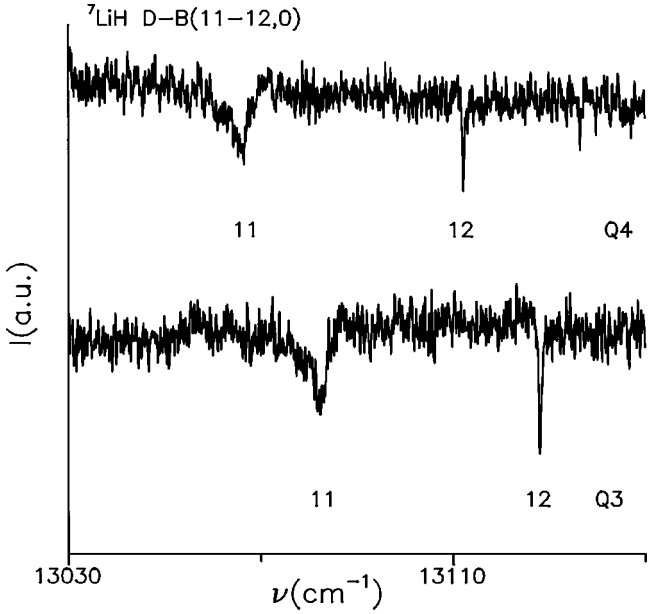


FIG. 7. OODR spectra. Two broad spectral lines represent the D-B(11,0), Q(3), and Q(4) transitions, and two narrow ones the D-B(12, 0), Q(3), and Q(4) transitions.

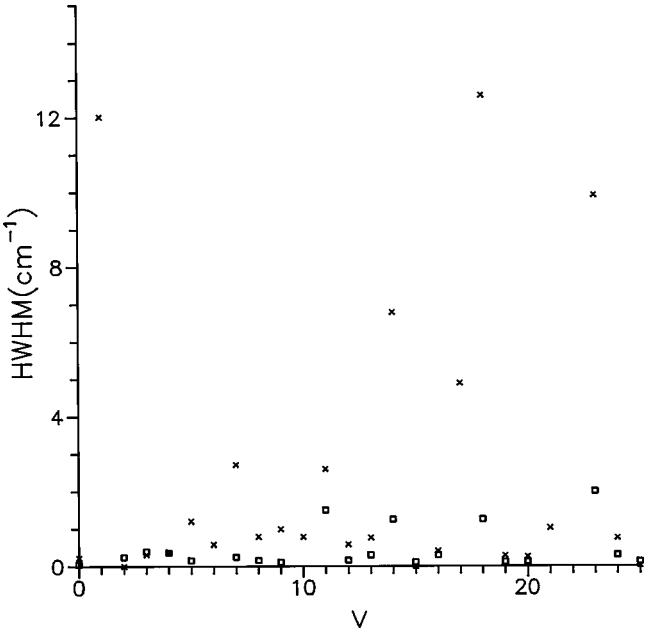


FIG. 8. Plot of observed spectral linewidths (open square symbol) and calculated nonradiative resonance widths (cross symbol) [HWHM in cm^{-1}] with respect to the vibrational quantum number.

With either the adiabatic potential presented in Table I or that reported previously by Boutalib and Gadéa,[24] the results of our intensity calculation indicate that the Franck–Condon factors (or transition moments) are separately 0.7 (0.2) for the $D\,^1\Sigma^+ - B\,^1\Pi\,(1,0)$ band and 0.04 (0.01) for the D-B(6,0) band, but negligible for the five bands between the five outer well (0, 2–5) vibrational levels of the ^{7}LiH $D\,^1\Sigma^+$ state and the $B\,^1\Pi(0)$ vibrational level. For the $D\,^1\Sigma^+ - A\,^1\Sigma^+$ case, the Franck–Condon factors (or transition moments) are also negligible for the D-A(0–5, 6–9) vibrational bands, but quite appreciable for the D-A(6,6–9) bands. Thus some strong OODR transitions among the D-B(1,0) band and some detectable transitions among the D-A(6, 6–9) and D-B(6,0) bands are expected to be observed if the adiabatic Born–Oppenheimer approach is indeed adequate for the *D* state.

Among those vibrational levels observed via the intermediate $B\,^1\Pi(0)$ vibrational level, the lowest four observed vibrational levels are already attributed to the outer well vibrational levels and they are assigned as the $v=0$, 2–4 levels (see also Table III), not expected from the result of an intensity calculation. On the other hand, we did not observe any OODR transition which might connect to the $v=1$ level, although the D-B(1,0) band was expected to be very strong. Among those vibrational levels observed via the intermediate $A\,^1\Sigma^+(6-9)$ vibrational levels, the lowest three levels are assigned as the $v=3-5$ levels, not expected again from the result of an intensity calculation. Besides, a vibrational level lying at $\sim$47 996 cm^{-1} has a rotational constant of 0.95 $\pm 0.15\,cm^{-1}$, larger than any other observed rotational constant, and it is assigned as the $v=7$ level, not the $v=6$ level as expected adiabatically. These discrepant observed results indicate that the real spectroscopic characteristics of the *D* state could not be mimicked by the Franck–Condon factors

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(or transition moments) calculated only with adiabatic potentials, and that a better approach should be followed.

In summary, every measure of above three different aspects, the energetics, the spectral line shape and the transition probability, favors the vibronic approach to characterize the LiH $D\,^1\Sigma^+$ state, as was advocated by Gemperle and Gadéa.[27]

IV. CONCLUSIONS

The ^{7}LiH $D\,^1\Sigma^+$ excited electronic state has been observed for the first time by a pulsed optical–optical double resonance fluorescence depletion spectroscopic technique. Several rovibrational levels of the $A\,^1\Sigma^+$ and $B\,^1\Pi$ electronic states were used as the intermediate states. Among 22 vibrational levels, 128 rovibrational levels have been observed. The spectral assignment is identified by the observed rotational structures and term values and by a comparison between the derived vibrational and rotational constants with the theoretical values. An *ab initio* adiabatic potential energy curve for the $D\,^1\Sigma^+$ state and relevant $D\,^1\Sigma^+$–$A\,^1\Sigma^+$ and $D\,^1\Sigma^+$–$B\,^1\Pi$ transition moment functions are presented. An overall measure of three different aspects, the energetics, the spectral line shape, and the transition probability, strongly favors the argument that the vibronic approach is a better approach for characterizing the excited $D\,^1\Sigma^+$ electronic state.

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